

AI-assisted Stochastic Optimization for GPU Data Centers Lifecycle Planning

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Abstract

The recent advancement in artificial intelligence (AI) boosts the demand for AI-related workloads, significantly increasing GPU deployment and associated power consumption in data centers. This trend accelerates hardware upgrades, introducing the need for GPU upgrades planning, reallocation, and retirement. Existing lifecycle planning methods typically rely on static provisioning or heuristic-based strategies, which make it challenging to address uncertainties in workload demand, hardware performance degradation, power limitations, etc. As a result, poor resource utilization and increased operating costs are common in data centers.

We propose an AI-assisted stochastic optimization framework for GPU lifecycle planning in data centers. This framework utilizes large language models to generate predictive scenarios for future workload demands, hardware performance decay, etc., in different cases. These AI-generated insights are integrated into a stochastic optimization model, which helps stakeholders make decisions about the timing of upgrades and retirements of GPU and cooling systems in data centers.

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1 Introduction

Recent growth of large language models (LLMs) and other AI-driven services led to a rapid growth in GPU-based data centers. However, it introduces a key challenge: determining how to manage data center facilities, upgrade and retire GPUs to fulfill performance demand while minimizing costs under power and cooling constraints. Traditional approaches often use fixed replacement cycles or heuristic decisions [1, 3], leading to over-provisioning or under-provisioning

as workloads evolve. Demand for AI workloads can be highly variable, and hardware performance can degrade non-linearly based on our observation, highlighting the limits of current planning methods.

Previous work applies machine learning (ML) techniques for data center infrastructure management, demand forecasting, and real-time resource adjustments [2, 4, 6]. Although these methods improve predictive accuracy and operational responsiveness, they have not been fully integrated into long-range lifecycle decisions such as the timing and scale of GPU purchases or retirements. Traditional stochastic optimization has been used to handle uncertainties in resource management [5, 7], but most existing work assumes simplified scenario generation. They have not investigated the uncertainties in AI workload growth, hardware performance decay patterns, and hardware market fluctuations. This gap between forecasting and multiperiod decision-making remains a significant barrier to efficient GPU lifecycle management.

In this paper, we propose an AI-assisted approach that integrates LLMs for scenario generation with a stochastic optimization framework. The LLM-based scenario generation technique constructs diverse future states for workloads, GPU performance, and cost parameters, reflecting possible technology releases and market changes. Our stochastic optimization model then determines how to purchase and retire GPUs under localized power and cooling capacity constraints to reduce the cost for data center management. Our contributions can be concluded to:

- We present a LLM scenario generation pipeline that captures the possibilities of uncertainties in demand, hardware depreciation, and power/cooling limits.
- We formulate a stochastic optimization problem that investigates hardware purchase, retirement, and the option to upgrade cooling systems.
- Based on the simulation results, our method outperforms the current heuristic upgrade method in all slow, linear, and fast growth scenarios.

2 Background

2.1 LLM Workloads on the Rise

The AI workloads, especially large transformer-based models used in generative AI, are skyrocketing. Industry analysts believe AI data center capacity must increase annually by at least 20–30% in

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the coming years, which means significantly higher GPU hours and compute requirements each quarter. It's challenging for the current GPU data centers to meet future demand without a well-considered upgrade strategy. In the intuition, the data center should adapt its hardware to serve increasingly large models and higher query volumes. Even with model optimizations, LLM inference and training are computationally intensive, introducing the challenges of existing infrastructures.

2.2 Data Center Infrastructure Constraints

Limited Power Capacity: The facility can supply only a fixed amount of power to GPU racks, capping the number of high-wattage devices that can run simultaneously. Each V100 or A100 draws hundreds of watts; adding more GPUs or higher-TDP models risks overloading the power budget. Therefore, any increase in computing must come with better FLOPs per Watt efficiency rather than simply more power draw. Newer GPUs usually have greater performance per watt, which gives the potential for GPU upgrade.

Cooling Limitations: The existing racks use air cooling, sufficient for 300–400W GPUs but challenged by the heat generation of newer 700W devices. High-power GPUs like the H100 in SXM form factor (700W TDP), cannot be densely deployed with standard air cooling, as they would overheat or cause performance drop. Upgrading to liquid cooling in the data center is a consideration to accommodate H100s at full performance. However, this upgrade has a one-time capital cost and operational changes. The trade-off is that liquid cooling would allow higher rack densities and possibly improved energy efficiency to support the next generation of accelerators.

Hardware Depreciation & Residual Value: GPUs are expensive equipment, but they don't retain value for long in the fast-moving AI market. The resale values of older GPUs decay over time, especially as newer generations become available quickly. The current V100s, now two generations old, still have value on secondary markets. However, A100s will likewise lose value after H100s and BL100s are announced. This caused the intuition to retire older GPUs earlier to get funds that can make new purchases. On the other hand, running hardware to its full useful life can maximize the return on investment. Hence, retirement timing is a balancing act between performance needs and residual value recovery.

2.3 GPU performance study

To have a better understanding of computing to power performance of different generations of GPUs, we compare NVIDIA's V100, A100, and H100 GPUs in terms of performance and suitability for large AI workloads. Table 1 summarizes key specifications and metrics. The H100's Hopper architecture introduces faster memory (HBM3) and more compute units, delivering significantly higher throughput. And H100 supports 8-bit floating point (FP8) calculations and a Transformer Engine, enabling up to four times performance on AI workloads at lower precision than A100. Compared to the V100, the H100 has 8.3 times more computing power and 2.3 times higher rated power, making it approximately 3.6 times more power-efficient.

Table 1: Evolution of NVIDIA data center GPUs

Model	Year	Max TDP	Memory Bandwidth	Memory (GB)	FP16 (TFLOPS)
V100	2017	300W	900 GB/s	32	120
A100	2020	400W	2039 GB/s	80	312
H100	2022	700W	3000 GB/s	80	1000

3 Problem formulation

We formulate the GPU lifecycle planning as a stochastic optimization problem. The objective is to minimize the expected total lifecycle cost over a planning horizon of T quarters, in which we can purchase, retire, or continue operating any GPUs in $\mathcal{G}$.

3.1 Decision Variables

For each time period t (where $t = 1, 2, \dots, T$) and each GPU type $i \in \mathcal{G}$:

- $x_t^i \in \mathbb{Z}_{\geq 0}$: Number of new GPUs of type i purchased at period t .
- $r_t^i \in \mathbb{Z}_{\geq 0}$: Number of GPUs of type i retired at period t .
- $g_t^i \in \mathbb{Z}_{\geq 0}$: can be 0 Total operational GPUs of type i at the end of period t .

In addition, $y_t \in \{0, 1\}$ indicate whether a one-time cooling upgrade is performed at time t . The multiple cooling upgrades scenario will be addressed in future work.

3.2 Objective Function

We seek to minimize the *expected total cost* over the horizon, comprising purchase costs, cooling upgrade costs, operating (electricity) costs, minus any salvage (resale) value recovered by retiring GPUs:

$$\min_{x, r, t} \mathbb{E} \left[\sum_{t=1}^T \left(c_t^p + c_t^c + c_t^o - c_t^r \right) \right], \quad (1)$$

where c^p is the GPU purchase cost, c^c is the data center cooling cost, c^o is the data center operation, management cost, and c^r is the old GPU resale income.

3.3 Constraints

1) *Demand Fulfillment (Stochastic)*. Let $d_{t,s}$ be the required GPU performance in scenario s at time t , and $p_{t,s}^i$ be the effective performance of one GPU of type i in scenario s at time t .

$$\sum_{i \in \mathcal{G}} g_t^i p_t^i \geq d_t, \quad \forall t. \quad (2)$$

2) *GPU Inventory Balance*. For each $i \in \mathcal{G}$ and period t :

$$g_t^i = g_{t-1}^i + x_t^i - r_t^i, \quad \forall i; t = 1, \dots, T, \quad (3)$$

g_0^i is given by the initial state, x_t^i and r_t^i are nonnegative integers.

3) *Power/Cooling Constraint*. Let w^i be the power requirement for a GPU of type i , and $w_{\max}^{\text{air}}$, $w_{\max}^{\text{liquid}}$ be the data center power limits under air cooling vs. upgraded cooling. Then for each t :

$$\sum_{i \in \mathcal{G}} g_t^i w^i \leq w_{\max}^{\text{air}} + y_t (w_{\max}^{\text{liquid}} - w_{\max}^{\text{air}}), \quad (4)$$

where $y_t = 1$ means the cooling upgrade has been installed by period t , enabling the higher power capacity.

4) *Cooling Upgrade Constraint.* One time cooling upgrade:

$$\sum_{t=1}^T y_t \leq 1, \quad (5)$$

$$y_{t+1} \geq y_t, \quad \forall t = 1, \dots, T-1. \quad (6)$$

Constraint (6) ensures that once the upgrade is done, all subsequent periods remain upgraded.

5) *Data Center Power Limit.* In addition to the GPUs' own power usage, the cooling infrastructure itself consumes power. w_c is the power used by cooling. Define w_{all} as the overall power limit of the data center. Then for each time period t , we add:

$$\sum_{i \in \mathcal{G}} g_t^i w^i + w_c^t \leq w_{\text{all}}, \quad (7)$$

where w_c^t depends on whether we have upgraded the cooling system, this constraint ensures that the GPU power and the appropriate cooling power never exceed the data center's power capacity.

6) *Non-negativity and Integrality.* All GPU-related decision variables must be nonnegative integers:

$$x_t^i, r_t^i, g_t^i \in \mathbb{Z}_+ \quad \text{and} \quad y_t \in \{0, 1\}, \quad \forall i, \forall t. \quad (8)$$

3.4 Cost Components

Purchase Cost. c_p^i is the cost of buying a single GPU of type i .

$$c_t^p = \sum_{i \in \mathcal{G}} c_p^i x_t^i. \quad (9)$$

Cooling Upgrade. c_c is the one-time cost of liquid-cooling retrofit:

$$c_t^c = c_c(y_t - y_{t-1}). \quad (10)$$

Operational Cost. $c_o(t)$ is the operational cost per unit power in period t , which is a linear function of the GPU number, resource usage, and people management cost. h_{quarter} is the number of hours in one quarter:

$$c_t^o = c_o(t) h_{\text{quarter}} \sum_{i \in \mathcal{G}} g_t^i w^i. \quad (11)$$

Resale Value. Let $v_{\text{res}}^i(t)$ be the per-GPU residual value for a type- i GPU retired at time t . Then:

$$c_t^r = \sum_{i \in \mathcal{G}} r_t^i v_{\text{res}}^i(t). \quad (12)$$

3.5 Stochastic Parameters

In our model, the following quantities are subject to uncertainty:

Workload Demand, d_t : The total required GPU performance at time t . For example, d_t could vary due to changes in user adoption, LLM service expansions, or seasonal spikes.

GPU Performance Degradation, p_t^i : Each GPU type i might deliver slightly different effective performance over time (e.g., caused by partial hardware aging or driver inefficiencies).

Residual (Salvage) Value, $v_{\text{res}}^i(t)$: The per-GPU resale value at time t , which might fluctuate with market conditions.

To capture these uncertainties, we add random fluctuations (e.g., Gaussian or lognormal noise) around these baseline values within each scenario. Thus, each scenario provides a *mean trajectory* for demand, GPU cost, and performance. In this way, the decision-maker can explore how robust the planning solution is to variability within each scenario. After introducing stochastic noise within each scenario, the original problem becomes:

$$\min \sum_{\omega \in \Omega_s} p_{\omega} \sum_{t=1}^T \left(c_{t,\omega}^p + c_t^c + c_{t,\omega}^o - c_{t,\omega}^r \right), \quad (13)$$

where Ω is the set of stochastic realizations (e.g., Gaussian noise perturbations) associated with this scenario. The parameters $c_{t,\omega}^p$, $c_{t,\omega}^o$, and $c_{t,\omega}^r$ incorporate uncertainty in demand, performance, and resale values. The optimization seeks to minimize the total expected cost across both macro scenarios and their associated stochastic realizations.

4 LLM-Assisted Scenario Generation

Data center planning spans different potential futures, from rapid exponential growth in LLM workloads to more moderate or even stable demand. Conventionally, data center operators manually enumerate relevant scenarios based on budget, commercial plans, and market conditions, relying on experience. This process can be both time-consuming and prone to bias. Instead, an LLM can *accelerate scenario generation* by proposing distinct, coherent narratives—each describing a different combination of demand trends, technology breakthroughs, and market dynamics. While we rely on the LLM for creativity and domain-inspired plausibility, we retain full control over the underlying stochastic parameters (e.g., setting the distribution type and variance) used in each scenario.

4.1 Workflow

The LLM-based scenario generation framework proceeds as:

Identify Macro Cases. We first decide which broad classes of futures we want to study: for example, *High Demand Growth*, *Slow GPU Releases*, *Strong Secondary Market*, etc.

Prompt the LLM. Provide structured prompts describing the context (e.g., time horizon, current data center status, potential technological events) and ask the LLM to produce scenario descriptors.

Curate LLM Output. Review the LLM's textual responses and extract the key scenario parameters (e.g., baseline demand for each quarter, presumed release dates for new GPUs). Each scenario thus has a coherent storyline that sets the mean or nominal values.

Tune Stochastic Distributions. For each scenario, define explicit noise models for $d_{t,\omega}$, $p_{t,\omega}^i$, and $v_{\text{res}}^i(t, \omega)$. For instance, if the LLM indicates "Demand grows 20% per quarter," it can be modeled as a lognormal random variable with mean μ (corresponding to 20% growth) and variance σ^2 , capturing short-term fluctuations.

4.2 Example

Suppose we solicit the following scenario from the LLM:

“Over 2025–2027, LLM services expand aggressively, doubling demand yearly. NVIDIA’s next-gen GPU arrives in mid-2026, older GPU resale values down by 40%. Electricity costs rise slowly, by 5% per year.”

From this description, we extract baseline: (i) demand grows 20–25% each quarter, (ii) new GPU releases occur in Q2 2026, and (iii) older GPU resale values drop abruptly around Q2–Q3 2026. Then, for each quarter t , we define random variables—e.g., $d_{t,\omega} \sim \mathcal{N}(\mu_t, \sigma^2)$ —where μ_t follows the LLM’s stated 20–25% growth pattern, and σ^2 is set by our domain knowledge or stress-testing strategy (e.g., a 10% standard deviation around the mean). We repeat this for GPU performance and resale value if needed. With AI-assisted scenario generation, we can easily explore a massive number of what-if studies that cover different future demands and market trends.

5 Evaluation

We evaluate our stochastic optimization approach by simulating a mid-sized data center’s GPU lifecycle planning.

AI-Assisted Stochastic Planning: Our method uses the optimization framework described in Section 3 with scenarios derived from LLM-generated scenarios and noise.

Heuristic Upgrade Strategy: Upgrades are triggered by simple thresholds (e.g., “Buy new GPUs if current utilization exceeds 90%”), and cooling upgrades occur only when power limits are exceeded for two consecutive quarters.

5.1 Simulation Setup

Table 2 summarizes the key input parameters for the simulation. We generate three different demand growth trends (*High*, *Moderate*, *Low*). Within each scenario, we sample Ω_s realizations from Gaussian distributions around that scenario’s baseline for workload demand, GPU resale values, and operational cost.

Table 2: Key Simulation Parameters

Parameter	Value (Example)
Time Horizon	12 quarters (3 years)
GPU Types	{V100, A100, H100}
Purchase Costs	A100 = \$10k, H100 = \$20k
Power Caps	Air: 3 kW; Liquid: 12 kW
Cooling Upgrade Cost	\$50k (one-time)
Scenarios (S)	High, Medium, Low Demand Growth
Micro Draws (Ω_s)	10 draws/scenario for random demand
Electricity Rate	\$0.10/kWh (varies $\pm 10\%$ in micro noise)
Total power	2MW

5.2 Metrics and Results

We measure *total life-cycle cost*: purchases + cooling upgrade + electricity expense – salvage. Figure 1 compares the proposed method versus the baseline under three distinct demand growth scenarios.

In the *Fast-Growth* scenario, our method achieves a total cost of \$261M compared to \$357M for the manual approach, yielding a savings of about 27%. Under *Linear-Growth* demand, the cost difference is \$204M vs. \$282M (28% savings). For *Slow-Growth* demand, the costs are \$109M vs. \$186M, saving up to 40%.

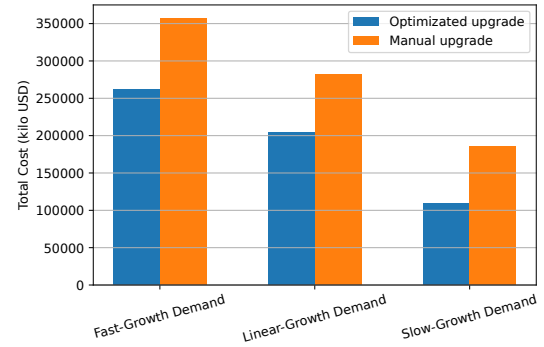


Figure 1: Illustrative total life-cycle cost under the proposed method vs. the manual heuristic.

5.3 Discussion

Our results indicate that incorporating stochastic optimization with AI-assisted scenario generation introduces more cost-effective decisions. In future work, we will address more complex situations to study the overall optimized strategies among different scenarios.

6 Conclusion

In this paper, we propose an AI-assisted stochastic optimization approach that effectively balances data center facilities’ upgrade and retirement under uncertainties. By integrating LLM-generated scenarios, we demonstrate the potential of AI-assisted data center management. Preliminary results show that this method outperforms current threshold-based upgrade strategies by reducing the datacenter upgrade cost by up to 40%.

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